

Master MVA

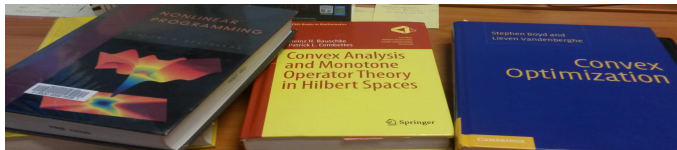
Optimization Reminders

Part I

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Reference books



- ▶ **D. Bertsekas**, Nonlinear programming, Athena Scientific, Belmont, Massachussets, 1996.
- ▶ **Y. Nesterov**, Introductory Lectures on Convex Optimization : A Basic Course, Springer, 2004.
- ▶ **S. Boyd and L. Vandenberghe**, Convex optimization, Cambridge University Press, 2004.
- ▶ **H. H. Bauschke and P. L. Combettes**, Convex Analysis and Monotone Operator Theory in Hilbert Spaces, Springer, New York, 2017.

A few mathematical background

Hilbert spaces

A (real) **pre-Hilbertian space** $\mathcal{H}$ is a real vector space endowed with a function $\langle \cdot | \cdot \rangle : \mathcal{H}^2 \rightarrow \mathbb{R}$, called **inner product** such that

- ① $(\forall (x, y) \in \mathcal{H}^2) \langle x | y \rangle = \langle y | x \rangle$
- ② $(\forall (x, y, z) \in \mathcal{H}^3) \langle x + y | z \rangle = \langle x | z \rangle + \langle y | z \rangle$
- ③ $(\forall (x, y) \in \mathcal{H}^2) (\forall \alpha \in \mathbb{R}) \langle \alpha x | y \rangle = \alpha \langle x | y \rangle$
- ④ $(\forall x \in \mathcal{H}) \langle x | x \rangle \geq 0$ and
 $\langle x | x \rangle = 0 \Leftrightarrow x = 0.$

► The associated norm is

$$(\forall x \in \mathcal{H}) \quad \|x\| = \sqrt{\langle x | x \rangle}.$$

Hilbert spaces

A (real) Hilbert space $\mathcal{H}$ is a complete pre-Hilbertian space.

▶ Particular cases

▶ $\mathcal{H} = \mathbb{R}^N$ (Euclidean space with dimension N)

▶ $\mathcal{H} = L^2(\mathbb{R})$ (square summable functions on $\mathbb{R}$).

Hilbert spaces

Let $\mathcal{H}$ and $\mathcal{G}$ be two Hilbert spaces.

A linear operator $L: \mathcal{H} \rightarrow \mathcal{G}$ is **bounded** if

$$\|L\| = \sup_{\|x\|_{\mathcal{H}} \leq 1} \|Lx\|_{\mathcal{G}} < +\infty$$

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- ▶ A linear operator from $\mathcal{H}$ to $\mathcal{G}$ is continuous if and only if it is bounded.
- ▶ In finite dimension, every linear operator is bounded.
- ▶ Unless otherwise specified, it will be assumed that all the underlying Hilbert spaces are finite dimensional.

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$\mathcal{B}(\mathcal{H}, \mathcal{G})$: Banach space of (bounded) linear operators from $\mathcal{H}$ to $\mathcal{G}$.

Hilbert spaces

Let $\mathcal{H}$ and $\mathcal{G}$ be two Hilbert spaces.

Let $L \in \mathcal{B}(\mathcal{H}, \mathcal{G})$. Its **adjoint L^*** is the operator in $\mathcal{B}(\mathcal{G}, \mathcal{H})$ defined as

$$(\forall (x, y) \in \mathcal{H} \times \mathcal{G}) \quad \langle y \mid Lx \rangle_{\mathcal{G}} = \langle L^*y \mid x \rangle_{\mathcal{H}}.$$

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Example 1 :

$$\text{If} \quad L: \mathcal{H} \rightarrow \mathcal{H}^n: x \mapsto (x, \dots, x)$$

$$\text{then} \quad L^*: \mathcal{H}^n \rightarrow \mathcal{H}: y = (y_1, \dots, y_n) \mapsto \sum_{i=1}^n y_i$$

Proof :

$$\langle Lx \mid y \rangle = \langle (x, \dots, x) \mid (y_1, \dots, y_n) \rangle = \sum_{i=1}^n \langle x \mid y_i \rangle = \left\langle x \mid \sum_{i=1}^n y_i \right\rangle$$

Hilbert spaces

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$$(\forall (x, y) \in \mathcal{H} \times \mathcal{G}) \quad \langle Lx \mid y \rangle = \langle x \mid L^*y \rangle.$$

Example 2 : $\mathcal{H} = \mathbb{R}^{N_1 \times N_2}$, $\mathcal{G} = \mathbb{R}^{M_1 \times M_2}$, equipped with Frobenius norm, and

$$(\forall X \in \mathcal{H}) \quad LX = L_1 X L_2^\top$$

where $L_1 \in \mathbb{R}^{M_1 \times N_1}$ and $L_2 \in \mathbb{R}^{M_2 \times N_2}$. Then

$$(\forall Y \in \mathcal{G}) \quad L^*Y = L_1^\top Y L_2.$$

$$\begin{aligned} \text{Proof : } \langle LX \mid Y \rangle &= \text{tr}(LX Y^\top) = \text{tr}(L_1 X L_2^\top Y^\top) \\ &= \text{tr}(X L_2^\top Y^\top L_1) = \langle X \mid L_1^\top Y L_2 \rangle. \end{aligned}$$

Hilbert spaces

Let $\mathcal{H}$ and $\mathcal{G}$ be two Hilbert spaces.

Let $L \in \mathcal{B}(\mathcal{H}, \mathcal{G})$. Its **adjoint L^*** is the operator in $\mathcal{B}(\mathcal{G}, \mathcal{H})$ defined as

$$(\forall (x, y) \in \mathcal{H} \times \mathcal{G}) \quad \langle Lx \mid y \rangle = \langle x \mid L^*y \rangle.$$

- ▶ We have $\|L^*\| = \|L\|$.
- ▶ If L is bijective (i.e. an **isomorphism**) then $L^{-1} \in \mathcal{B}(\mathcal{G}, \mathcal{H})$ and $(L^{-1})^* = (L^*)^{-1}$.
- ▶ If $\mathcal{H} = \mathbb{R}^N$ and $\mathcal{G} = \mathbb{R}^M$ then $L^* = L^\top$.

Infinite valued functions

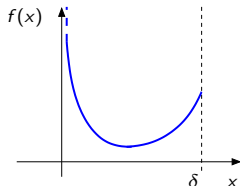
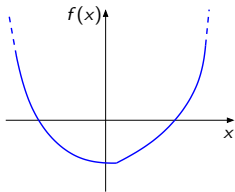
Functional analysis : definitions

Let S be a nonempty set of a Hilbert space $\mathcal{H}$.

Let $f: S \rightarrow]-\infty, +\infty]$.

- ▶ The **domain** of f is $\text{dom } f = \{x \in S \mid f(x) < +\infty\}$.
- ▶ The function f is **proper** if $\text{dom } f \neq \emptyset$.

Domains of the functions ?



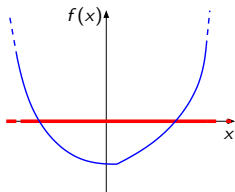
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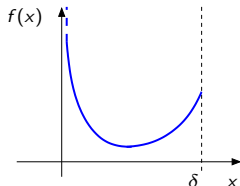
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Domains of the functions ?



$\text{dom } f = \mathbb{R}$
(proper)



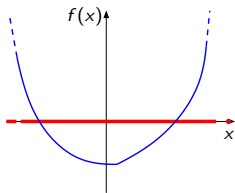
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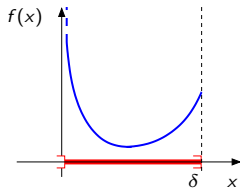
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Domains of the functions ?



$\text{dom } f = \mathbb{R}$
(proper)



$\text{dom } f =]0, \delta]$
(proper)

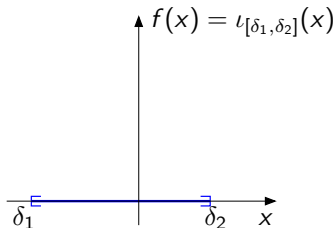
Functional analysis : definitions

Let $C \subset \mathcal{H}$.

The indicator function of C is

$$(\forall x \in \mathcal{H}) \quad \iota_C(x) = \begin{cases} 0 & \text{if } x \in C \\ +\infty & \text{otherwise.} \end{cases}$$

Example : $C = [\delta_1, \delta_2]$



Limits inf and sup

Let $(\xi_n)_{n \in \mathbb{N}}$ be a sequence of elements in $[-\infty, +\infty]$.

Its **infimum limit** is $\liminf \xi_n = \lim_{n \rightarrow +\infty} \inf \{ \xi_k \mid k \geq n \} \in [-\infty, +\infty]$

and its **supremum limit** is $\limsup \xi_n = \lim_{n \rightarrow +\infty} \sup \{ \xi_k \mid k \geq n \} \in [-\infty, +\infty]$.

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- ▶ $\limsup \xi_n = -\liminf(-\xi_n)$
- ▶ $\liminf \xi_n \leq \limsup \xi_n$
- ▶ $\lim_{n \rightarrow +\infty} \xi_n = \bar{\xi} \in [-\infty, +\infty]$ if and only if $\liminf \xi_n = \limsup \xi_n = \bar{\xi}$.

Epigraph

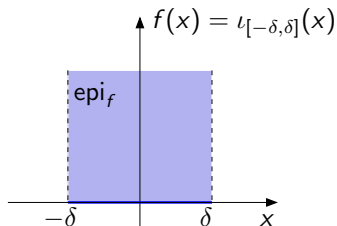
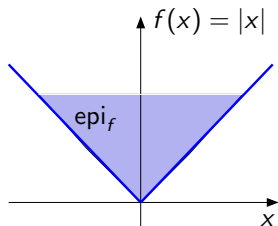
Let $f : \mathcal{H} \rightarrow]-\infty, +\infty]$. The **epigraph** of f is

$$\text{epi } f = \{(x, \zeta) \in \text{dom } f \times \mathbb{R} \mid f(x) \leq \zeta\}$$

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Lower semi-continuity

Let $f : \mathcal{H} \rightarrow]-\infty, +\infty]$.

f is a **lower semi-continuous** (l.s.c.) function at $x \in \mathcal{H}$ if, for every sequence $(x_n)_{n \in \mathbb{N}}$ of $\mathcal{H}$,

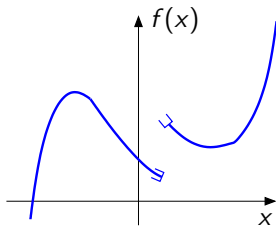
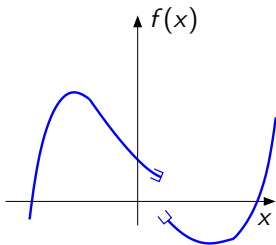
$$x_n \rightarrow x \quad \Rightarrow \quad \liminf f(x_n) \geq f(x).$$

Lower semi-continuity

Let $f : \mathcal{H} \rightarrow]-\infty, +\infty]$.

f is a lower semi-continuous function on $\mathcal{H}$ if and only if $\text{epi } f$ is closed

► l.s.c. functions?

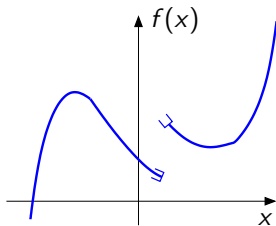
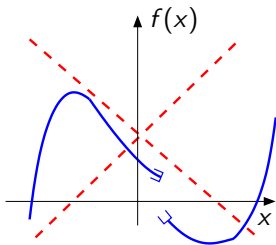


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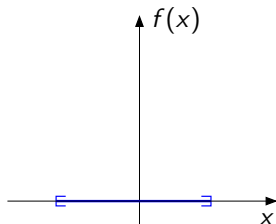
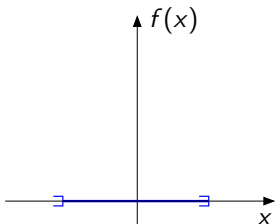


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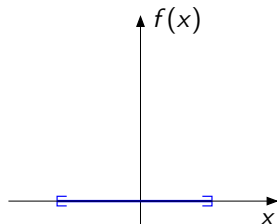
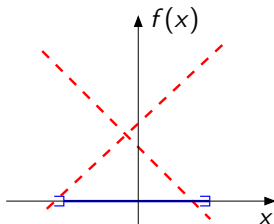


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Let $f : \mathcal{H} \rightarrow]-\infty, +\infty]$.

f is a lower semi-continuous function on $\mathcal{H}$ if and only if $\text{epi } f$ is closed

► l.s.c. functions?



Lower semi-continuity

- ▶ Every continuous function on $\mathcal{H}$ is l.s.c.
- ▶ Every finite sum of l.s.c. functions is l.s.c.
- ▶ Let $(f_i)_{i \in I}$ be a family of l.s.c functions.
Then, $\sup_{i \in I} f_i$ is l.s.c.

Does a minimum exist ?

Minimizers

Let S be a nonempty set of a Hilbert space $\mathcal{H}$.

Let $f: S \rightarrow]-\infty, +\infty]$ be a proper function and let $\hat{x} \in S$.

- ▶ $\hat{x}$ is a **local minimizer** of f if $\hat{x} \in \text{dom } f$ and there exists an open neighborhood O of $\hat{x}$ such that

$$(\forall x \in O \cap S) \quad f(\hat{x}) \leq f(x).$$

- ▶ $\hat{x}$ is a **(global) minimizer** of f if

$$(\forall x \in S) \quad f(\hat{x}) \leq f(x).$$

Minimizers

Let S be a nonempty set of a Hilbert space $\mathcal{H}$.

Let $f: S \rightarrow]-\infty, +\infty]$ be a proper function and let $\hat{x} \in S$.

- ▶ $\hat{x}$ is a strict local minimizer of f if there exists an open neighborhood O of $\hat{x}$ such that

$$(\forall x \in (O \cap S) \setminus \{\hat{x}\}) \quad f(\hat{x}) < f(x).$$

- ▶ $\hat{x}$ is a strict (global and unique) minimizer of f if

$$(\forall x \in S \setminus \{\hat{x}\}) \quad f(\hat{x}) < f(x).$$

Existence of a minimizer

Weierstrass theorem

Let S be a nonempty compact set of a Hilbert space $\mathcal{H}$.

Let $f : S \rightarrow]-\infty, +\infty]$ be a proper l.s.c function.

Then, there exists $\hat{x} \in S$ such that

$$f(\hat{x}) = \inf_{x \in S} f(x).$$

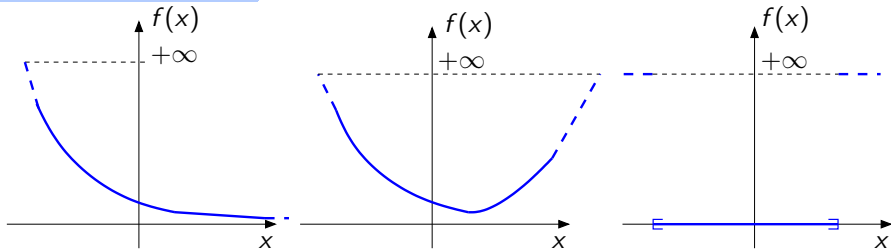
Existence of a minimizer

Let $\mathcal{H}$ be a Hilbert space. Let $f: \mathcal{H} \rightarrow]-\infty, +\infty]$.
 f is **coercive** if $\lim_{\|x\| \rightarrow +\infty} f(x) = +\infty$.

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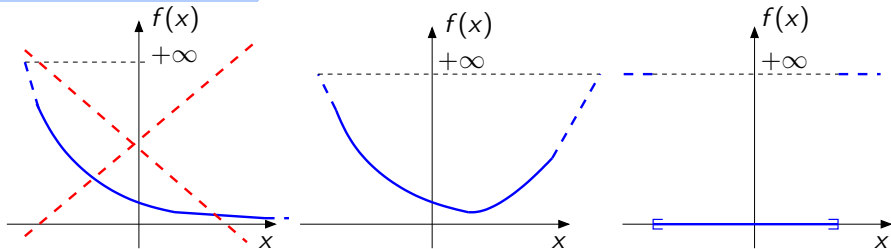
Coercive functions?



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Coercive functions?



Existence of a minimizer

Theorem

Let $\mathcal{H}$ be a (finite dimensional) Hilbert space.

Let $f : \mathcal{H} \rightarrow]-\infty, +\infty]$ be a proper l.s.c. coercive function.

Then, the set of minimizers of f is a nonempty compact set.

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Then, the set of minimizers of f is a nonempty compact set.

Proof : Since f is proper, there exists $x_0 \in \mathcal{H}$ such that $f(x_0) \in \mathbb{R}$. The coercivity of f implies that there exists $\eta \in]0, +\infty[$ such that, for every $x \in \mathcal{H}$ satisfying $\|x - x_0\| > \eta$, $f(x) > f(x_0)$.

Let $S = \{x \in \mathcal{H} \mid \|x - x_0\| \leq \eta\}$, $S \cap \text{dom } f \neq \emptyset$ and S is compact.

Then, there exists $\hat{x} \in S$ such that $f(\hat{x}) = \inf_{x \in S} f(x) \leq f(x_0)$. Thus, $f(\hat{x}) = \inf_{x \in \mathcal{H}} f(x)$.

$\text{Argmin } f \subset S$ is bounded. In addition, if $(x_n)_{n \in \mathbb{N}}$ is a sequence of minimizers converging to $\bar{x} \in \mathcal{H}$. Then, $f(\bar{x}) \leq \liminf f(x_n) = \inf_{x \in \mathcal{H}} f(x)$ and, consequently, $\bar{x} \in \text{Argmin } f$. Therefore, $\text{Argmin } f$ is closed.

Exercise 1

Let $\varphi:]0, +\infty[\rightarrow \mathbb{R}$ be a continuous function such that $\lim_{\xi \rightarrow 0} \varphi(\xi) = 0$ and let f be the function defined as

$$f(x) = \begin{cases} \sum_{i=1}^N \varphi(x^{(i)}) & \text{if } x = (x^{(i)})_{1 \leq i \leq N} \in]0, +\infty[^N \\ +\infty & \text{if } \begin{cases} (\exists j \in \{1, \dots, N\}) \ x^{(j)} < 0 \\ (\forall i \in \{1, \dots, N\}) \ x^{(i)} \neq 0. \end{cases} \end{cases}$$

1. How can we extend the definition of function f so that it is l.s.c. on $\mathbb{R}^N$?
2. What can be said about the existence of a minimizer of this function on a nonempty closed subset of the set $C = \{(x^{(i)})_{1 \leq i \leq N} \in [0, +\infty[^N \mid \sum_{i=1}^N x^{(i)} = 1\}$?

Hints from differential calculus

Necessary condition for the existence of a minimizer (Euler's inequality)

Theorem

Let D be an open subset of a Hilbert space $\mathcal{H}$ and let $C \subset D$. Let $f: D \rightarrow]-\infty, +\infty]$ be differentiable at $\hat{x} \in C$. If $\hat{x}$ is a local minimizer of f on C then, for every $y \in \mathcal{H}$ such that $[\hat{x}, y] \subset C$,

$$\langle \nabla f(\hat{x}) \mid y - \hat{x} \rangle \geq 0.$$

If $\hat{x} \in \text{int}(C)$, then the condition reduces to

$$\nabla f(\hat{x}) = 0.$$

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Remark : A zero of the gradient ∇f is called a **critical point** (or stationary point) of f .

Necessary condition for the existence of a minimizer

Theorem

Let C be an open subset of a Hilbert space $\mathcal{H}$. Let $f: C \rightarrow \mathbb{R}$ be differentiable on C . Let f be twice differentiable at $\hat{x} \in C$. If $\hat{x}$ is a local minimizer of f on C , then

- (i) $\nabla f(\hat{x}) = 0$;
- (ii) the Hessian $\nabla^2 f(\hat{x})$ of f at $\hat{x}$ is positive semi-definite, i.e.

$$(\forall z \in \mathcal{H}) \quad \langle z \mid \nabla^2 f(\hat{x})z \rangle \geq 0.$$

Sufficient conditions for the existence of a minimizer

Theorem

Let C be an open subset of a Hilbert space $\mathcal{H}$. Let $f: C \rightarrow \mathbb{R}$ be differentiable on C .

- (i) If f is twice differentiable at $\hat{x} \in C$, $\nabla f(\hat{x}) = 0$ and the Hessian $\nabla^2 f(\hat{x})$ of f at $\hat{x}$ is positive definite, i.e.

$$(\forall z \in \mathcal{H} \setminus \{0\}) \quad \langle z \mid \nabla^2 f(\hat{x})z \rangle > 0.$$

then f has a strict local minimum at $\hat{x}$.

Sufficient conditions for the existence of a minimizer

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then f has a strict local minimum at $\hat{x}$.

- (ii) If f is twice differentiable on an open neighborhood $D \subset C$ of $\hat{x}$, $\nabla f(\hat{x}) = 0$ and the Hessian of f is positive semi-definite on D , i.e.

$$(\forall x \in D)(\forall z \in \mathcal{H}) \quad \langle z \mid \nabla^2 f(x)z \rangle \geq 0,$$

then f has a local minimum at $\hat{x}$.

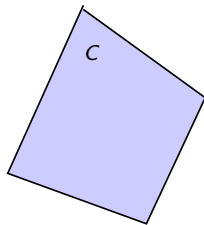
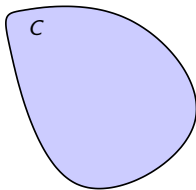
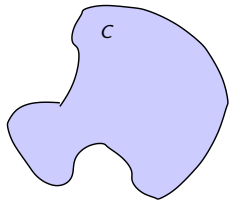
Magic of convexity

Convex set : definition

Let $\mathcal{H}$ be a Hilbert space. $C \subset \mathcal{H}$ is a **convex set** if

$$(\forall (x, y) \in C^2)(\forall \alpha \in]0, 1[) \quad \alpha x + (1 - \alpha)y \in C$$

Convex sets ?

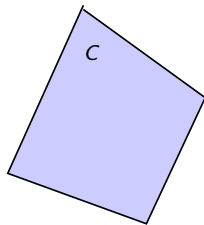
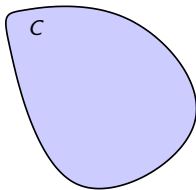
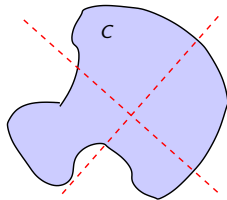


Convex set : definition

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$$(\forall (x, y) \in C^2)(\forall \alpha \in]0, 1[) \quad \alpha x + (1 - \alpha)y \in C$$

Convex sets ?



Convex set : properties

- ▶ $\emptyset$ is considered as a convex set.
- ▶ If C is a convex set, then $(\forall n \in \mathbb{N}^*) (\forall (x_1, \dots, x_n) \in C^n)$
 $(\forall (\alpha_1, \dots, \alpha_n) \in [0, +\infty[^n)$ with $\sum_{i=1}^n \alpha_i = 1$,

$$\sum_{i=1}^n \alpha_i x_i \in C.$$

- ▶ Every vector (affine) space is convex.
- ▶ If C is a convex set, then $\text{int}(C)$ and $\overline{C}$ are convex sets.

Convex set : properties

- ▶ If C is a convex set then, for every $\alpha \in \mathbb{R}$,

$$\alpha C = \{\alpha x \mid x \in C\}$$

is a convex set.

- ▶ If C_1 and C_2 are convex sets, then

$$C_1 \times C_2$$

$$C_1 + C_2 = \{x_1 + x_2 \mid (x_1, x_2) \in C_1 \times C_2\}$$

are convex sets.

- ▶ If $(C_i)_{i \in I}$ is a family of convex sets of $\mathcal{H}$, then $\bigcap_{i \in I} C_i$ is convex.

Convex hull

Let $\mathcal{H}$ be a Hilbert space and $C \subset \mathcal{H}$. The **convex hull** of C is the smallest convex set including C . It is denoted by $\text{conv}(C)$.

- ▶ $\text{conv}(C)$ is the intersection of all the convex sets including C .
- ▶ Let $x \in \mathcal{H}$. $x \in \text{conv}(C)$ if and only if $(\exists n \in \mathbb{N}^*) (\exists (x_1, \dots, x_n) \in C^n)$
 $(\exists (\alpha_1, \dots, \alpha_n) \in]0, +\infty[^n \text{ with } \sum_{i=1}^n \alpha_i = 1 \text{ such that}$

$$x = \sum_{i=1}^n \alpha_i x_i.$$

Convex function : definitions

$f : \mathcal{H} \rightarrow]-\infty, +\infty]$ is a **convex function** if

$$(\forall (x, y) \in \mathcal{H}^2)(\forall \alpha \in]0, 1[)$$

$$f(\alpha x + (1 - \alpha)y) \leq \alpha f(x) + (1 - \alpha)f(y)$$

Convex function : definitions

$f : \mathcal{H} \rightarrow]-\infty, +\infty]$ is a **convex function** if

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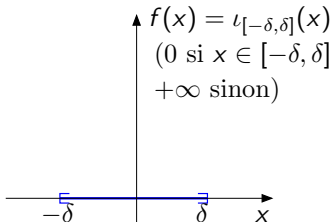
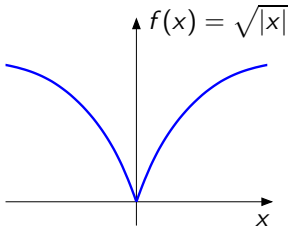
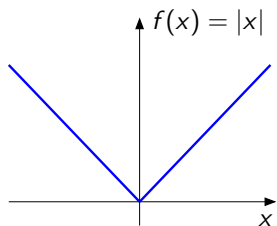
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Convex functions ?



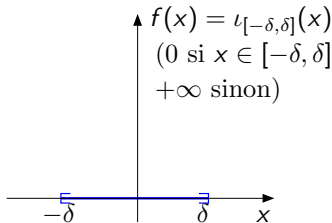
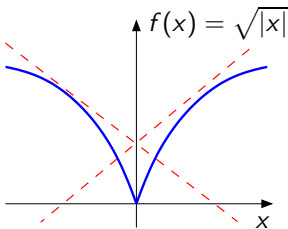
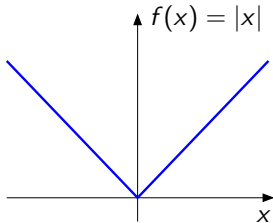
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Convex functions ?



Convex function : definitions

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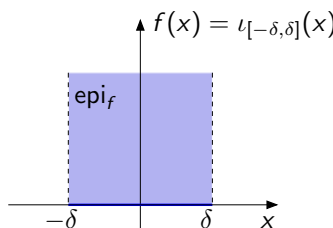
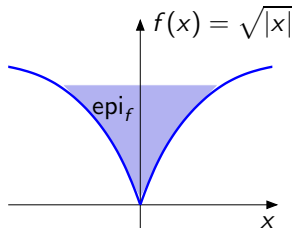
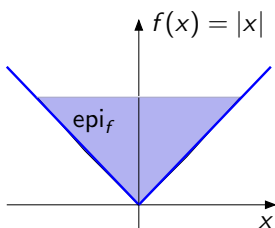
► $f : \mathcal{H} \rightarrow]-\infty, +\infty]$ is concave if $-f$ is convex.

Convex functions : definition

$f : \mathcal{H} \rightarrow]-\infty, +\infty]$ is convex $\Leftrightarrow$ its epigraph is a convex set.

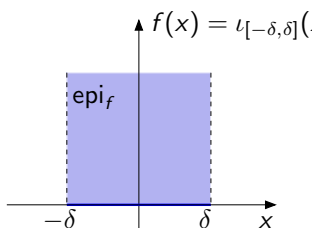
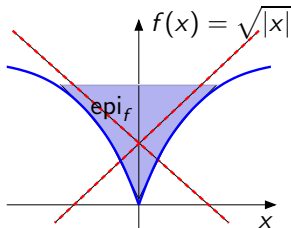
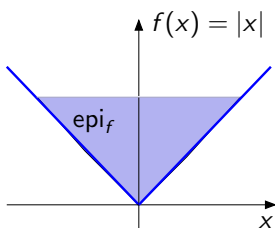
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Convex functions : properties

- ▶ If $f: \mathcal{H} \rightarrow]-\infty, +\infty]$ is convex, then $\text{dom } f$ is convex and its lower level set at height $\eta \in \mathbb{R}$

$$\text{lev}_{\leq \eta} f = \{x \in \mathcal{H} \mid f(x) \leq \eta\}$$

is a convex set.

- ▶ $f: \mathcal{H} \rightarrow]-\infty, +\infty]$ is convex if and only if $(\forall (x, y) \in (\text{dom } f)^2)$
 $\varphi_{x,y}: [0, 1] \rightarrow]-\infty, +\infty]: \alpha \mapsto f(\alpha x + (1 - \alpha)y)$ is convex.

Convex functions : properties

- ▶ Every finite sum of convex functions is convex.
- ▶ Let $(f_i)_{i \in I}$ be a family of convex functions. Then, $\sup_{i \in I} f_i$ is convex.
- ▶ $\Gamma_0(\mathcal{H})$: class of convex, l.s.c., and proper functions from $\mathcal{H}$ to $] -\infty, +\infty]$.
- ▶ Let $C \subset \mathcal{H}$.
 $\iota_C \in \Gamma_0(\mathcal{H}) \Leftrightarrow C$ is a nonempty closed convex set.
Proof : $\text{epi}_{\iota_C} = C \times [0, +\infty[$.

Strictly convex functions

Let $\mathcal{H}$ be a Hilbert space. Let $f: \mathcal{H} \rightarrow]-\infty, +\infty]$.

f is **strictly convex** if

$$(\forall x \in \text{dom } f)(\forall y \in \text{dom } f)(\forall \alpha \in]0, 1[)$$

$$x \neq y \quad \Rightarrow \quad f(\alpha x + (1 - \alpha)y) < \alpha f(x) + (1 - \alpha)f(y).$$

Strictly convex functions

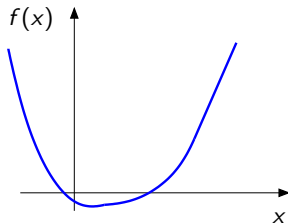
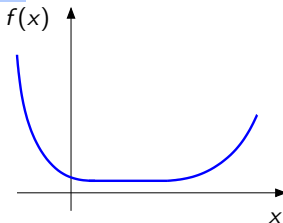
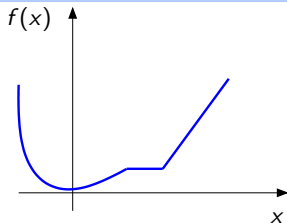
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Strictly convex functions?



Strictly convex functions

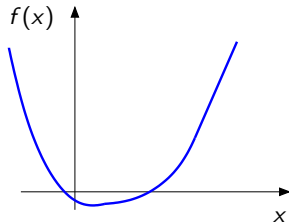
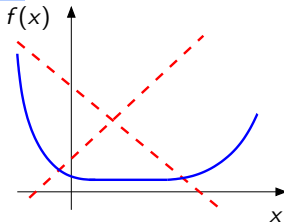
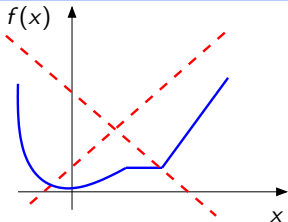
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$$x \neq y \Rightarrow f(\alpha x + (1 - \alpha)y) < \alpha f(x) + (1 - \alpha)f(y).$$

Strictly convex functions?



Minimizers of a convex function

Theorem

Let $\mathcal{H}$ be a Hilbert space. Let $f: \mathcal{H} \rightarrow]-\infty, +\infty]$ be a proper convex function such that $\mu = \inf f > -\infty$.

- ▶ $\{x \in \mathcal{H} \mid f(x) = \mu\}$ is convex.
- ▶ Every local minimizer of f is a global minimizer.
- ▶ If f is strictly convex, then there exists at most one minimizer.

Minimizers of a convex function

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- ▶ Every local minimizer of f is a global minimizer.
- ▶ If f is strictly convex, then there exists at most one minimizer.

Proof : Let $\Omega = \{x \in \mathcal{H} \mid f(x) = \mu\}$. Let $(x, y) \in \Omega^2$ and let $\alpha \in [0, 1]$. We have

$$f(\alpha x + (1 - \alpha)y) \leq \alpha f(x) + (1 - \alpha)f(y) = \mu$$

which shows that $\alpha x + (1 - \alpha)y \in \Omega$.

Minimizers of a convex function

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- ▶ $\{x \in \mathcal{H} \mid f(x) = \mu\}$ is convex.
- ▶ Every local minimizer of f is a global minimizer.
- ▶ If f is strictly convex, then there exists at most one minimizer.

Proof : Let $\hat{x}$ be a local minimizer of f . For every $y \in \mathcal{H} \setminus \{\hat{x}\}$, there exists $\alpha \in]0, 1[$ such that

$$\begin{aligned} f(\hat{x}) &\leq f(\hat{x} + \alpha(y - \hat{x})) \leq (1 - \alpha)f(\hat{x}) + \alpha f(y) \\ \Rightarrow f(\hat{x}) &\leq f(y) \end{aligned}$$

If f is strictly convex, the inequality is strict.

Existence and uniqueness of a minimizer

Theorem

Let $\mathcal{H}$ be a Hilbert space and C a closed convex subset of $\mathcal{H}$. Let $f \in \Gamma_0(\mathcal{H})$ such that $\text{dom } f \cap C \neq \emptyset$.

If f is coercive or C is bounded, then there exists $\hat{x} \in C$ such that

$$f(\hat{x}) = \inf_{x \in C} f(x).$$

If, moreover, f is strictly convex, this minimizer $\hat{x}$ is unique.

Exercise 2

Let $\mathcal{H}$ be a Hilbert space.

1. Show that the function $x \mapsto \|x\|^2$ is strictly convex.
2. A function $f: \mathcal{H} \rightarrow]-\infty, +\infty]$ is β -convex with $\beta \in \mathbb{R}$ if there exists a convex function $g: \mathcal{H} \rightarrow]-\infty, +\infty]$ such that

$$f = g + \frac{\beta}{2} \|\cdot\|^2.$$

If $\beta \geq 0$, then f is *strongly convex* and, if $\beta \leq 0$, then f is *weakly convex*. Show that every strongly convex function is strictly convex.

3. Show that a function $f: \mathcal{H} \rightarrow]-\infty, +\infty]$ is β -convex if and only if

$$(\forall (x, y) \in \mathcal{H}^2)(\forall \alpha \in]0, 1[)$$

$$f(\alpha x + (1 - \alpha)y) + \alpha(1 - \alpha)\frac{\beta}{2}\|x - y\|^2 \leq \alpha f(x) + (1 - \alpha)f(y).$$

Convex + smooth

Characterization of differentiable convex functions

Let $f: \mathcal{H} \rightarrow]-\infty, +\infty]$ be differentiable on $\text{dom } f$, which is a nonempty open convex set.

Then, f is convex if and only if

$$(\forall (x, y) \in (\text{dom } f)^2) \quad f(y) \geq f(x) + \langle \nabla f(x) \mid y - x \rangle.$$

Characterization of differentiable strictly convex functions

Let $f: \mathcal{H} \rightarrow]-\infty, +\infty]$ be differentiable on $\text{dom } f$, which is a nonempty open convex set.

Then, f is strictly convex if and only if, for every $(x, y) \in (\text{dom } f)^2$ with $x \neq y$,

$$f(y) > f(x) + \langle \nabla f(x) \mid y - x \rangle.$$

Characterization of differentiable convex functions

Let $f: \mathcal{H} \rightarrow]-\infty, +\infty]$ be differentiable on $\text{dom } f$, which is a nonempty open convex set.

Then, f is convex if and only if ∇f is monotone on $\text{dom } f$, i.e.

$$(\forall (x, y) \in (\text{dom } f)^2) \quad \langle \nabla f(y) - \nabla f(x) \mid y - x \rangle \geq 0.$$

Characterization of strictly differentiable convex functions

Let $f: \mathcal{H} \rightarrow]-\infty, +\infty]$ be differentiable on $\text{dom } f$, which is a nonempty open convex set.

Then, f is strictly convex if and only if ∇f is strictly monotone on $\text{dom } f$, i.e. for every $(x, y) \in (\text{dom } f)^2$ with $x \neq y$,

$$\langle \nabla f(y) - \nabla f(x) \mid y - x \rangle > 0.$$

Characterization of twice differentiable convex functions

Let $\mathcal{H}$ be a Hilbert space.

Let $f: \mathcal{H} \rightarrow]-\infty, +\infty]$ be a twice differentiable function on $\text{dom } f$, which is a nonempty open convex set.

- ▶ f is convex if and only if, for every $x \in \text{dom } f$, $\nabla^2 f(x)$ is positive semi-definite.
- ▶ If, for every $x \in \text{dom } f$, $\nabla^2 f(x)$ is positive definite, then f is strictly convex.

Condition for the existence of a minimizer

Theorem

Let $\mathcal{H}$ be Hilbert space.

Let $f: \mathcal{H} \rightarrow]-\infty, +\infty]$ be a differentiable convex function on $\text{dom } f$, which is an open set. Let $C \subset \text{dom } f$ be a nonempty convex set. $\hat{x} \in C$ is a (global) minimizer of f on C if and only if

$$(\forall y \in C) \quad \langle \nabla f(\hat{x}) \mid y - \hat{x} \rangle \geq 0.$$

If $\hat{x} \in \text{int}(C)$, then the condition reduces to

$$\nabla f(\hat{x}) = 0.$$

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If $\hat{x} \in \text{int}(C)$, then the condition reduces to

$$\nabla f(\hat{x}) = 0.$$

Proof : We have already seen that the inequality is a necessary condition for $\hat{x}$ to be a local minimizer of f on C and that it reduces to the vanishing condition on the gradient if $\hat{x} \in \text{int}(C)$.

Condition for the existence of a minimizer

Theorem

Let $\mathcal{H}$ be Hilbert space.

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If $\hat{x} \in \text{int}(C)$, then the condition reduces to

$$\nabla f(\hat{x}) = 0.$$

Proof : Conversely, assume that the inequality holds. Let $y \in C$. Since f is convex and differentiable,

$$f(y) \geq f(\hat{x}) + \langle \nabla f(\hat{x}) \mid y - \hat{x} \rangle \geq f(\hat{x}).$$

Hence, $\hat{x}$ is a minimizer of f on C .

Exercise 3

Let $f: \mathcal{H} \rightarrow \mathbb{R}$ be differentiable.

Show that f is β -convex with $\beta \in \mathbb{R}$ if and only if

$$(\forall (x, y) \in \mathcal{H}^2) \quad f(y) \geq f(x) + \langle \nabla f(x) \mid y - x \rangle + \frac{\beta}{2} \|y - x\|^2.$$

Exercise 4

Let

$$f: \mathbb{R}^N \rightarrow \mathbb{R}$$
$$(x^{(i)})_{1 \leq i \leq N} \mapsto \ln \left(\sum_{i=1}^N \exp(x^{(i)}) \right).$$

Show that f is convex. Is it strictly convex?

Projections

Projection onto a closed convex set

Theorem

Let C be a nonempty closed convex set of a Hilbert space $\mathcal{H}$.

- (i) For every $x \in \mathcal{H}$, there exists a unique point $\hat{x}$ in C which lies at minimum distance of x . The application $P_C: \mathcal{H} \rightarrow C$ which maps every $x \in \mathcal{H}$ to its associated point $\hat{x}$ is called the projection onto C .
- (ii) For every $x \in \mathcal{H}$, $\hat{x} = P_C(x)$ if and only if $\hat{x} \in C$ and

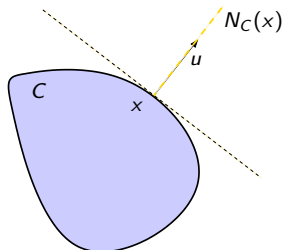
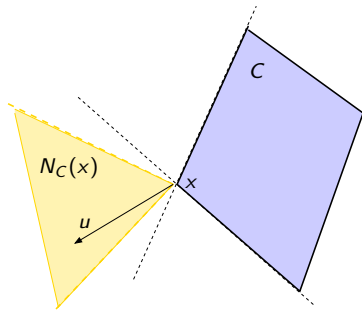
$$(\forall y \in C) \quad \langle x - \hat{x} \mid y - \hat{x} \rangle \leq 0.$$

Geometrical interpretation

Let C be a nonempty subset of $\mathcal{H}$.

For every $x \in \mathcal{H}$, the **normal cone** to C at x is defined as

$$N_C(x) = \begin{cases} \{u \in \mathcal{H} \mid (\forall y \in C) \langle u \mid y - x \rangle \leq 0\} & \text{if } x \in C \\ \emptyset & \text{otherwise.} \end{cases}$$



Geometrical interpretation

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For every $x \in \mathcal{H}$, the **normal cone** to C at x is defined as

$$N_C(x) = \begin{cases} \{u \in \mathcal{H} \mid (\forall y \in C) \langle u \mid y - x \rangle \leq 0\} & \text{if } x \in C \\ \emptyset & \text{otherwise.} \end{cases}$$

- ▶ $(\forall x \in C)$, $N_C(x)$ is closed and convex
- ▶ If $x \in \text{int } C$, then $N_C(x) = \{0\}$.
- ▶ If C is a vector space, then for every $x \in C$, $N_C(x) = C^\perp$.
- ▶ Let C be nonempty closed convex set of a Hilbert space $\mathcal{H}$. For every $x \in \mathcal{H}$,

$$\hat{x} = P_C(x) \quad \Leftrightarrow \quad x - \hat{x} \in N_C(\hat{x}).$$

Examples of projections

- ▶ If C is a (closed) vector space of a Hilbert space $\mathcal{H}$, then P_C is the (linear) orthogonal projection onto C . Then, for every $x \in \mathcal{H}$,

$$\hat{x} = P_C(x) \quad \Leftrightarrow \quad \begin{cases} \hat{x} \in C \\ x - \hat{x} \in C^\perp. \end{cases}$$

Examples of projections

- Let $\mathcal{H}$ be a Hilbert space, let $u \in \mathcal{H} \setminus \{0\}$, let $\delta \in \mathbb{R}$, and let C be the closed half space defined as

$$C = \{x \in \mathcal{H} \mid \langle u \mid x \rangle \leq \delta\}.$$

The projection onto C is

$$(\forall x \in \mathcal{H}) \quad P_C(x) = \begin{cases} x + \frac{\delta - \langle u \mid x \rangle}{\|u\|^2} u & \text{if } \langle u \mid x \rangle > \delta \\ x & \text{otherwise.} \end{cases}$$

Properties of the projection

- ▶ Let C be a nonempty closed convex set of a Hilbert space $\mathcal{H}$. The projection onto C is a **firmly nonexpansive operator**, i.e.

$$(\forall (x, y) \in \mathcal{H}^2) \quad \|P_C(x) - P_C(y)\|^2 \leq \langle x - y \mid P_C(x) - P_C(y) \rangle.$$

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- ▶ The projection onto C is a **nonexpansive operator**, i.e.

$$(\forall (x, y) \in \mathcal{H}^2) \quad \|P_C(x) - P_C(y)\| \leq \|x - y\|.$$

- ▶ The projection onto C is uniformly continuous.
- ▶ The **distance to C** defined as

$$(\forall x \in \mathcal{H}) \quad d_C(x) = \inf_{y \in C} \|x - y\| = \|x - P_C(x)\|$$

is continuous.

Exercise 5

Let D be a nonempty closed convex set of $\mathbb{R}^N$.

Let $L \in \mathbb{R}^{N \times N}$ be an orthogonal matrix and let

$$C = \{x \in \mathbb{R}^N \mid Lx \in D\}.$$

1. Show that the projection onto C is well-defined.
2. Show that $P_C = L^\top \circ P_D \circ L$.
3. Express P_C when $D = [0, +\infty[^N$.